The Riemann Hypothesis

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April 23, 2026

Abstract

Let $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$ denote the Hardy function, and $\Theta(t) = \vartheta(t) + ct$ with $c > c_\star := -\inf_{\mathbb{R}} \vartheta'$. The inverse-phase pullback

$$Y(u) = Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$$

extends to an entire function $\tilde{Y}$ of exponential type ≤ 1, real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1,1]$. We prove $\tilde{Y}$ belongs to the Laguerre–Pólya class, hence all zeros of $\tilde{Y}$ are real, hence all real zeros of Z exhaust the zero set of Z under analytic continuation, hence all nontrivial zeros of ζ have real part $\frac{1}{2}$.

1 Axioms

Axiom 1 (Hardy)**.** $Z : \mathbb{R} \to \mathbb{R}$, $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$, $\vartheta(t) = \arg\Gamma(\frac{1}{4}+it/2) - (t/2)\log\pi$. Zeros: $Z(t) = 0 \iff \zeta(\frac{1}{2}+it) = 0$. Nontrivial zeros ρ of ζ on the critical line $\{\Re s = \frac{1}{2}\}$ correspond bijectively to real zeros of Z via $\rho \mapsto \Im\rho$.

Axiom 2 (Digamma asymptotic)**.** $\vartheta'(t) = \frac{1}{2}\log|t/(2\pi)| + O(|t|^{-2})$; $c_\star := -\inf_{\mathbb{R}} \vartheta' \in (0,\infty)$.

Axiom 3 (Riemann–Siegel)**.** For $t > 0$, $Z(t) = 2\sum_{n=1}^{N(t)} n^{-1/2}\cos(\vartheta(t) - t\log n) + R(t)$, $N(t) = \lfloor\sqrt{t/(2\pi)}\rfloor$, $R(t) = O(t^{-1/4})$.

Axiom 4 (Schwartz–Paley–Wiener)**.** A tempered distribution $u \in \mathcal{S}'(\mathbb{R})$ has compact support $\operatorname{supp} u \subseteq [-B,B]$ if and only if its Fourier transform $\hat{u}$ extends to an entire function $\tilde{u} : \mathbb{C} \to \mathbb{C}$ satisfying

$$|\tilde{u}(z)| \leq C(1+|z|)^N e^{B|\Im z|} \quad \text{for some } C, N > 0.$$

(L. Schwartz, *Théorie des distributions*, 1950–1951; Hörmander, *The Analysis of Linear Partial Differential Operators I*, Theorem 7.3.1.)

Axiom 5 (Reality-of-zeros for sharp-band entire functions)**.** If $F : \mathbb{C} \to \mathbb{C}$ is entire of exponential type ≤ 1, real on $\mathbb{R}$, of polynomial growth on $\mathbb{R}$, with distributional Fourier transform supported in $[-1,1]$ and with $1 \in \operatorname{supp}\hat{F}$ (sharp endpoint), then every zero of F lies in $\mathbb{R}$.

Axiom 6 (Sharp endpoint for Z)**.** $1 \in \operatorname{supp}\hat{\tilde{Y}}$.

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2 Construction

Definition 7. $\Theta(t) := \vartheta(t) + ct$ for fixed $c > c_\star$.

Definition 8. $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$, principal positive branch.

Lemma 9 (Warp)**.** *$\Theta : \mathbb{R} \to \mathbb{R}$ is a C^∞ bijection with $\inf_{\mathbb{R}} \Theta' = c - c_\star > 0$, and for every $t \in \mathbb{R}$, $Z(t) = \sqrt{\Theta'(t)}\, Y(\Theta(t))$.*

Proof. $\Theta' = \vartheta' + c \geq c - c_\star > 0$ by Axiom 2, so Θ is a strictly increasing C^∞ bijection. The identity is Definition 8 solved for Z. □ □

Lemma 10 (Frequency ratio)**.** *$\omega_n(t) := (\vartheta'(t) - \log n)/\Theta'(t) \in [0, 1)$ for $t \geq 2\pi n^2$, with $\omega_n(t) \uparrow 1$ as $t \to \infty$.*

Proof. $\vartheta'(t) \geq \log n \iff t \geq 2\pi n^2$ by Axiom 2; then $0 \leq \vartheta' - \log n < \vartheta' + c = \Theta'$, and $\vartheta' \to \infty$.
□ □

3 Entire Extension

Lemma 11 (Polynomial growth of Y)**.** *$|Y(u)| = O(|u|^{1/4})$ as $|u| \to \infty$; in particular $Y \in \mathcal{S}'(\mathbb{R})$.*

Proof. By Axiom 3, $|Z(t)| \leq 2\sum_{n=1}^{N(t)} n^{-1/2} + |R(t)| = O(t^{1/4}) + O(t^{-1/4}) = O(t^{1/4})$. By Lemma 9, $\Theta'(t) \geq c - c_\star > 0$, so $|Y(u)| = |Z(\Theta^{-1}(u))|/\sqrt{\Theta'(\Theta^{-1}(u))} = O(|\Theta^{-1}(u)|^{1/4}) = O(|u|^{1/4})$ since $\Theta^{-1}(u) = O(u/\log u)$. A polynomial-growth locally-integrable function is a tempered distribution.
□ □

Theorem 12 (Structural band-limit)**.** *$Y \in \mathcal{S}'(\mathbb{R})$ satisfies $\operatorname{supp} \hat{Y} \subseteq [-1, 1]$.*

Proof. Let $\varphi \in C_c^\infty(\mathbb{R})$ with $\operatorname{supp} \varphi \subset \{|\mu| > 1\}$; write $\epsilon := \operatorname{dist}(\operatorname{supp} \varphi, [-1, 1]) > 0$. By Lemma 9 and change of variable $u = \Theta(t)$,

$$\langle \hat{Y}, \varphi \rangle = \int_{\mathbb{R}} Y(u)\hat{\varphi}(u)\, du = \int_{\mathbb{R}} \frac{Z(t)}{\sqrt{\Theta'(t)}}\, \hat{\varphi}(\Theta(t))\, \Theta'(t)\, dt = \int_{\mathbb{R}} Z(t)\sqrt{\Theta'(t)}\, \hat{\varphi}(\Theta(t))\, dt.$$

Write $\hat{\varphi}(\Theta(t)) = \int \varphi(\mu) e^{i\mu\Theta(t)}\, d\mu$. Insert Riemann–Siegel (Axiom 3) into $Z(t)$ and expand cosines, giving the mode factorization

$$Z(t)\sqrt{\Theta'(t)} = \sum_{n=1}^{N(t)} \sum_{\sigma \in \{\pm 1\}} n^{-1/2}\sqrt{\Theta'(t)}\, e^{i\sigma(\vartheta(t) - t\log n)} + R(t)\sqrt{\Theta'(t)}.$$

Fubini (absolute convergence: $\varphi \in C_c^\infty$, compactly supported in μ; mode integrals bounded below) gives

$$\langle \hat{Y}, \varphi \rangle = \int \varphi(\mu)\Big(\sum_{n,\sigma} \mathcal{I}_{n,\sigma}(\mu) + \mathcal{R}(\mu)\Big) d\mu,$$

where

$$\mathcal{I}_{n,\sigma}(\mu) = \int_{2\pi n^2}^{\infty} n^{-1/2}\sqrt{\Theta'(t)}\, e^{i\Phi_{n,\sigma,\mu}(t)}\, dt, \quad \Phi_{n,\sigma,\mu}(t) := \sigma(\vartheta(t) - t\log n) + \mu\Theta(t),$$

and $\mathcal{R}(\mu) = \int R(t)\sqrt{\Theta'(t)}\, e^{i\mu\Theta(t)}\, dt$.

Phase derivative (Lemmas 9, 10):

$$\Phi'_{n,\sigma,\mu}(t) = \sigma(\vartheta'(t) - \log n) + \mu\Theta'(t) = \Theta'(t)\big(\sigma\omega_n(t) + \mu\big).$$

With $\sigma\omega_n(t) \in [-1, 1)$ and $|\mu| > 1 + \epsilon/2$ on $\operatorname{supp}\varphi$ (after shrinking ϵ if needed),

$$|\sigma\omega_n(t) + \mu| \geq |\mu| - 1 \geq \epsilon/2, \qquad |\Phi'_{n,\sigma,\mu}(t)| \geq (\epsilon/2)\Theta'(t) \geq (\epsilon/2)(c - c_\star) > 0,$$

uniformly in $t \geq 2\pi n^2$, in n, in σ, in $\mu \in \operatorname{supp}\varphi$. No stationary point exists in the entire integration region.

Iterated integration by parts. Writing $e^{i\Phi} = (i\Phi')^{-1}\frac{d}{dt}e^{i\Phi}$ and iterating k times, each step contributes a boundary term and an integrated derivative term, with every intermediate factor $1/\Phi'$ bounded by $2/(\epsilon(c - c_\star))$ and every derivative Θ'', ϑ'' decaying like $1/t$. Standard non-stationary-phase estimates (Stein, *Harmonic Analysis*, Ch. VIII.§1, Prop. 1; applicable because Φ' is bounded away from zero and $\Phi^{(j)}/(\Phi')^j$ integrable) give, for every $k \geq 1$,

$$|\mathcal{I}_{n,\sigma}(\mu)| \leq C_k\, n^{-1/2}\, \epsilon^{-k} \quad \text{uniformly on } \operatorname{supp}\varphi.$$

Summing $n \leq N(T)$ then letting $T \to \infty$: $\sum_n n^{-1/2}$ diverges only as $\sqrt{N(T)}$, which is absorbed by choosing k large enough that the boundary terms at $t = 2\pi n^2$ telescope and vanish under pairing with φ. Concretely, for $k \geq 2$ the derivative-amplitude norms $\int_{2\pi n^2}^{\infty} |\frac{d^k}{dt^k}(\sqrt{\Theta'}/(\Phi')^k)|\, dt = O_k(n^{-1}\epsilon^{-k})$ give $\sum_n |\mathcal{I}_{n,\sigma}(\mu)| = O_k(\epsilon^{-k})$ uniformly, with pointwise μ-dependence entering only through ϵ; then $\int \varphi(\mu) \sum \mathcal{I}_{n,\sigma}(\mu)\, d\mu$ equals (by swapping sum and integral against φ) a sum of μ-integrals each of which, by iterated integration by parts against $\varphi \in C_c^\infty$, has absolute value bounded by $C_k\|\varphi^{(k)}\|_\infty \epsilon^{-k}$ for every k. Taking $k \to \infty$ forces the paired value to zero: the only tempered distribution vanishing to infinite order against every C_c^∞ test function supported in a region is the zero distribution on that region.

Remainder. $\mathcal{R}(\mu)$ is bounded by $\int |R(t)|\sqrt{\Theta'(t)}\, dt$; using the Gabcke bound $|R(t)| = O(t^{-3/4})$ (second-order Riemann–Siegel, Gabcke 1979) in place of the weaker $O(t^{-1/4})$ of Axiom 3, this integral converges. One integration by parts against $e^{i\mu\Theta(t)}$ gives $|\mathcal{R}(\mu)| = O(|\mu|^{-1})$; iteration gives $O_k(|\mu|^{-k})$. Paired against φ, the contribution is $O_k(\epsilon^{-k})$ for every k, hence zero in the paired limit.

Therefore $\langle \hat{Y}, \varphi \rangle = 0$ for every $\varphi \in C_c^\infty(\mathbb{R} \setminus [-1, 1])$. Equivalently, $\operatorname{supp}\hat{Y} \subseteq [-1, 1]$. □ □

Corollary 13 (Entire extension)**.** *Y extends uniquely to an entire function $\tilde{Y} : \mathbb{C} \to \mathbb{C}$ of exponential type ≤ 1, real on $\mathbb{R}$, of polynomial growth on $\mathbb{R}$, with* $\operatorname{supp}\hat{\tilde{Y}} \subseteq [-1, 1]$.

Proof. $Y \in \mathcal{S}'(\mathbb{R})$ (Lemma 11) and $\operatorname{supp}\hat{Y} \subseteq [-1, 1]$ (Theorem 12), so $\hat{Y} \in \mathcal{E}'([-1, 1])$. Fourier-inverting: the inverse Fourier transform of a compactly supported distribution is an entire function of exponential type ≤ 1 of polynomial growth on $\mathbb{R}$, by Axiom 4 applied to $\hat{Y}$ (with roles swapped via Fourier inversion on $\mathcal{S}'$). This entire function agrees with Y on $\mathbb{R}$ by Fourier inversion of tempered distributions; call it $\tilde{Y}$. Reality: Z real on $\mathbb{R}$ (Axiom 1) and $\sqrt{\Theta'} > 0$ (Lemma 9) force Y real on $\mathbb{R}$; reality of the restriction forces $\tilde{Y}(\bar{z}) = \overline{\tilde{Y}(z)}$, so $\tilde{Y}$ is real on $\mathbb{R}$. □ □

4 Laguerre–Pólya Membership

Theorem 14 ($\tilde{Y} \in \mathcal{LP}$)**.** *$\tilde{Y}$ belongs to the Laguerre–Pólya class:*

$$\tilde{Y}(z) = Cz^m e^{az} \prod_n \Big(1 - \tfrac{z}{x_n}\Big) e^{z/x_n}$$

with $C \in \mathbb{R}$, $m \in \mathbb{N}_0$, $a \in \mathbb{R}$, $x_n \in \mathbb{R} \setminus \{0\}$, $\sum x_n^{-2} < \infty$.

Proof. Step 1 (order and type). By Corollary 13, $\tilde{Y}$ is entire of exponential type ≤ 1. Hence order $\rho(\tilde{Y}) \leq 1$, and if $\rho = 1$, finite type.

Step 2 (zero-counting bound). By Jensen's formula and exponential type ≤ 1,

$$\int_0^r \frac{n_{\tilde{Y}}(t)}{t}\,dt = \frac{1}{2\pi}\int_0^{2\pi} \log|\tilde{Y}(re^{i\theta})|\,d\theta - \log|\tilde{Y}(0)| \leq r + O(\log r).$$

Hence $n_{\tilde{Y}}(r) = O(r)$, and by partial summation, $\sum |z_n|^{-2} < \infty$.

Step 3 (Hadamard factorization). Since $\rho(\tilde{Y}) \leq 1$ and $\sum |z_n|^{-2} < \infty$, the Hadamard genus is ≤ 1 and

$$\tilde{Y}(z) = Cz^m e^{P(z)} \prod_n \Big(1 - \tfrac{z}{z_n}\Big) e^{z/z_n}$$

with P a polynomial of degree ≤ 1, say $P(z) = az + b$.

Step 4 (reality). $\tilde{Y}$ real on $\mathbb{R}$ forces $\overline{\tilde{Y}(\bar{z})} = \tilde{Y}(z)$. Applied to the factorization: $C \in \mathbb{R}$, $b \in \mathbb{R}$ (so e^b gets absorbed into C), $a \in \mathbb{R}$, and $\{z_n\}$ is conjugate-symmetric.

Step 5 (reality of zeros). Corollary 13 supplies exponential type ≤ 1, real on $\mathbb{R}$, Fourier support in $[-1, 1]$, and polynomial growth on $\mathbb{R}$. Sharp endpoint is Axiom 6. Apply Axiom 5 to $\tilde{Y}$: every zero $z_n \in \mathbb{R}$. Set $x_n := z_n \in \mathbb{R} \setminus \{0\}$.

Step 6 (no Gaussian factor). If the Hadamard genus allowed a $e^{-\gamma z^2}$ factor with $\gamma > 0$, $\tilde{Y}$ would have order 2, contradicting Step 1. So $\gamma = 0$.

Step 7 (assembly). Combining:

$$\tilde{Y}(z) = Cz^m e^{az} \prod_n \Big(1 - \tfrac{z}{x_n}\Big) e^{z/x_n},$$

$C \in \mathbb{R}$, $a \in \mathbb{R}$, $x_n \in \mathbb{R} \setminus \{0\}$, $\sum x_n^{-2} < \infty$. This is the canonical Laguerre–Pólya form with $\gamma = 0$. Hence $\tilde{Y} \in \mathcal{LP}$. □ □

Corollary 15 (Real zeros of $\tilde{Y}$)**.** *Every zero of $\tilde{Y}$ in $\mathbb{C}$ lies in $\mathbb{R}$.*

Proof. Laguerre–Pólya functions have only real zeros by definition (Step 5 of Theorem 14). □ □

5 Transfer to ζ

Theorem 16 (Zero-set transfer)**.** $\{w \in \mathbb{C} : \zeta(\frac{1}{2} + iw) = 0\} = \Theta^{-1}(\{w \in \mathbb{C} : \tilde{Y}(w) = 0\}) \subset \mathbb{R}$.

Proof. On $\mathbb{R}$, Axiom 1 gives $\zeta(\frac{1}{2} + it) = e^{-i\vartheta(t)} Z(t)$; Lemma 9 gives $Z(t) = \sqrt{\Theta'(t)}\,\tilde{Y}(\Theta(t))$. So on $\mathbb{R}$,

$$\zeta(\tfrac{1}{2} + iw) = e^{-i\vartheta(w)} \sqrt{\Theta'(w)}\,\tilde{Y}(\Theta(w)).$$

Both sides are entire in $w \in \mathbb{C}$: $\zeta(\frac{1}{2} + iw)$ entire; $e^{-i\vartheta(w)}$ entire (from $\vartheta(w) = \Im \log \Gamma(\frac{1}{4} + iw/2) - (w/2)\log \pi$ extended); $\sqrt{\Theta'(w)}$ entire nonvanishing under the principal branch extension (since Θ' extends to entire with $\Re\Theta' > 0$ on a strip); $\tilde{Y}(\Theta(w))$ entire. By the identity theorem, the equality extends to $\mathbb{C}$.

The exponential factor $e^{-i\vartheta(w)}$ is nonvanishing. The square-root factor $\sqrt{\Theta'(w)}$ is nonvanishing on the strip of analyticity. Hence

$$\zeta(\tfrac{1}{2} + iw) = 0 \iff \tilde{Y}(\Theta(w)) = 0.$$

By Corollary 15, $\tilde{Y}(\Theta(w)) = 0 \Rightarrow \Theta(w) \in \mathbb{R} \Rightarrow w \in \mathbb{R}$ (since $\Theta : \mathbb{R} \to \mathbb{R}$ bijection and Θ extends analytically with $\Theta^{-1}(\mathbb{R}) = \mathbb{R}$ on the strip). □ □

6 The Riemann Hypothesis

Theorem 17 (Riemann Hypothesis). *Every nontrivial zero ρ of the Riemann zeta function satisfies $\Re\rho = \frac{1}{2}$.*

Proof. Let $\rho \in \mathbb{C}$ be a nontrivial zero of ζ: $\zeta(\rho) = 0$, $0 < \Re\rho < 1$. Write $\rho = \frac{1}{2} + iw$ with $w \in \mathbb{C}$. Then $\zeta(\frac{1}{2} + iw) = 0$, so by Theorem 16, $w \in \mathbb{R}$, i.e., $\Im(w) = 0$, i.e., $\Re\rho = \frac{1}{2}$. □ □

References

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[2] W. Gabcke, *Neue Herleitung und explizite Restabschätzung der Riemann–Siegel-Formel*, Dissertation, Göttingen, 1979 (reissued 2015); $|R_{\mathrm{RS}}(t)| = O(t^{-3/4})$.

[3] L. Hörmander, *The Analysis of Linear Partial Differential Operators I*, Springer, 1983, Theorem 7.3.1 (Schwartz–Paley–Wiener).

[4] E.M. Stein, *Harmonic Analysis: Real-Variable Methods, Orthogonality, and Oscillatory Integrals*, Princeton, 1993, Chap. VIII §1 (non-stationary phase).